

EOLLM Fusion Tasks — Top 20 (Final Shortlist for Team Review)

Synthesis of 5 independent model proposals (Claude, Grok, GPT, Gemini, Qwen, Kimi) + 4 internal adversarial agents. Ranked by your 3 criteria — **broad necessity**, **real usability**, **OSM-grounding** — plus the fusion bar.

FIXED SPEC (applies to every task)

- **Input:** always 4 street-view frames + 1 satellite image (red dot marks the point).
- **Answer:** 4 text labels (multiple choice). NO "pick a frame" answers. Where a task is directional, the labels name **what is there** ("toward the main road", "toward the dead-end", "toward the water"), NEVER the frame name ("cross_left"). This forces the model to *identify* the referent, not emit a frame index.
- **Fusion bar (acceptance):** a task qualifies iff **full-input accuracy** > **EACH single-view accuracy by our gate margin** — single view may be decent, it just must be clearly beaten: $\text{synergy} = \text{full} - \max(\text{sat}, \text{sv}) \geq +8 \text{ pts}$ AND $\text{fusion-win} = P(\text{full right} \wedge \text{both singles wrong}) \geq 15\%$ AND $\text{full} > \text{both singles}$. Single-view being *possible but insufficient* is exactly the target — graceful degradation, not chance-level. The gate decides per task before scaling.
- **City-balance (mandatory):** balance answer labels ~25/25/25/25 *within each city* to kill the satellite-reveals-city → modal-answer prior leak.

How to read this table

- **Conv.** = how many of the 5 external models independently proposed this family (consensus signal).
- **Label** = ground-truth source:  **Auto** (dense OSM geometry/tags + bearing, no vision) ·  **Auto-filter** + **vision-answer** (OSM selects candidates; VLM/human assigns the eye-level label).

TIER A — Build first. Highest necessity × usability × OSM-grounding, cleanest fusion.

1. Forward link form (route severance) ★ FLAGSHIP — consensus #1 (all 6)

Q: "Standing at the marked point, the map shows a road continuing in the direction you are facing. Physically, what is that onward route?" **Labels:** Drivable through-road · Pedestrian-only steps/stairs · Path or ramp (non-vehicular) · Dead-end / private (no public through-route). **Single-view but bad:** sat-only guesses "road" from the line (often wrong when it's stairs/gated); sv-only sees the surface but mis-reads cul-de-sac vs through. **Full needs both:** map's claimed continuation × street's physical form. **Label:**  Auto — **highway** class of the bearing-matched forward way + node-degree topology (**steps** → stairs; **footway/path/cycleway** → path; degree-1/ **noexit** → dead-end; road class → drivable). **Necessity:** THE routing/last-mile failure ("map says go, you can't"). Universal.

2. Forward crossing grade relation ★ — consensus #1 (all 6)

Q: "Following the road forward, it meets another road, railway, or waterway. How do they relate physically?" **Labels:** At-grade (same level) · Road passes OVER (bridge/flyover) · Road passes UNDER (tunnel/underpass) · No major crossing ahead. **Single-view but bad:** sat-only has NO Z-axis (over/under/at-grade look identical from above) → near-random among the three; sv-only sees a bridge/tunnel but can't bind it to the mapped crossing. **Full needs both:** map says "crosses X here" × street says "who's above whom." **Label:** 🟢 Auto — forward way ∩ (**waterway** / **railway** /major **highway**); **bridge**=* → over, **tunnel**=* / **layer**<0 → under, shared node → at-grade. Densely tagged (topology-critical). **Necessity:** Grade separation = primary cause of urban severance. Strongest "single view physically cannot" claim in the set.

3. Barrier crossing reachability (severance) — consensus 5/5 [directional → text labels]

Q: "A river, railway, or major road runs near the marked point. Where is the nearest crossing a pedestrian or cyclist can actually use?" **Labels:** Toward the forward road · Toward the road behind you · Toward the left cross-street · Toward the right cross-street. (*named by the network feature, gate-checked that a model can't read the label off frame geometry alone*) **Single-view but bad:** sat-only can't confirm a crossing permits pedestrians / sees no underpass; sv-only sees only foreground, misses a crossing around the corner. **Full needs both. Label:** 🟢 Auto — nearest **waterway** / **railway** / **motorway/trunk/primary** barrier; crossings = ways with **bridge** / **tunnel** / **ferry** or **footway/path/cycleway** spanning it (foot-access filter); bearing → label. **Necessity:** Severance + transport equity ("wrong side of the tracks"). Core walkability metric.

4. Filtered permeability / modal connectivity — consensus 4/5

Q: "The map shows the forward road continuing. What is its real connectivity for different travel modes?" **Labels:** Through-route for all modes · Vehicular dead-end but pedestrian/cycle through-route (filtered) · Absolute dead-end for all modes · Pedestrian/cycle-only by design. **Single-view but bad:** sat-only sees a continuous pavement ribbon (bollards sub-pixel) → over-predicts "through for all"; sv-only sees bollards but can't tell if it connects through beyond them. **Label:** 🟡 Auto-filter (**highway** topology + connectivity to a through-street) + vision for the bollard/gate state. **Necessity:** 15-minute-city / Low-Traffic-Neighbourhood core metric.

5. Onward route mode-and-continuity status — consensus 4/5 (GPT/Qwen "continuation status")

Q: "For the road that continues forward from the marked point, what is its combined status?" **Labels:** Vehicular through-connection · Pedestrian/cycle-only through-connection · Vehicular local access or dead-end · Pedestrian/cycle-only local access or dead-end. **Single-view but bad:** sat-only gets through-vs-dead-end (topology) but not mode (bollards invisible); sv-only gets mode (motorable vs stairs) but not through-vs-dead-end. **Each owns ONE bit → full needs both.** This 2×2 is the cleanest "each view supplies one independent bit" structure. **Label:** 🟡 Auto-filter (graph reachability + **highway** class) + vision for mode where untagged. **Necessity:** Navigation, emergency access, walkability continuity.

TIER B — Strong, build second. Slightly lower yield or needs a polygon re-query.

6. Misclassified-as-stairs detector (routing-bug) — consensus 4/5 [directional → text]

Q: "Among the routes the map shows leaving this junction (none of which the map labels as steps), which one is physically a staircase on the ground?" **Labels:** The forward route · The route behind · The left route · The right route. (text-named; the point is divergence: map says non-stairs, street shows stairs) **Single-view but bad:** sat-only can't see steps (under-tagged); sv-only "where are stairs" is partly solvable but can't tell which the map mis-labels. **Full needs both. Label:** 🟡 Auto-filter (candidate ways NOT tagged `steps`) + vision. ⚠️ low yield → mine hilly cities. **Necessity:** Famous "routed a wheelchair onto stairs" bug. High accessibility value.

7. Adjacent open-space access type — consensus 5/5 [recast directional → type labels]

Q: "The map shows a park / green / water space touching this point. What kind of access does its edge present here on the ground?" **Labels:** Open, walk-in edge or gateway · Gated/controlled entrance · Fenced / walled / sealed edge · No usable edge (private buildings between road and space). **Single-view but bad:** sat-only sees the polygon but not whether the edge is open vs walled; sv-only sees a fence/gate but doesn't know it's the mapped park or its extent. **Full needs both. Label:** 🟡 polygon (`leisure=park` / `recreation_ground` / `natural=water`) + "edge touched by a public `highway` " proxy + vision. **Needs polygon re-query. Necessity:** Green/blue-space access equity.

8. Street function (network role × eye-level regime) — consensus 4/5

Q: "Given how this street sits in the surrounding network and how it looks on the ground, what role does it actually play?" **Labels:** Through-traffic artery · Local access street · Shared / pedestrian-priority space · Service / back lane. **Single-view but bad:** sat-only knows network position, not regime; sv-only knows regime, not role. **They diverge** (pedestrianized arterial; minor rat-run) → full needs both. **Label:** 🟡 `highway` class + betweenness/degree + vision for calming/regime. ⚠️ class ↔ appearance correlates → oversample divergent cells, gate-verify. **Necessity:** Street-function / "stroad" / traffic calming policy.

9. Transverse barrier identity (rail vs water) — consensus 2/5

Q: "Two linear barriers — a railway AND a canal/river — run near this point. Which one does the visible crossing structure actually cross?" **Labels:** The railway · The waterway · Both have a crossing here · Neither (no usable crossing here). **Single-view but bad:** sat-only can't confirm a physical crossing at this point; sv-only — a narrow canal and single-track rail look near-identical under a bridge. **Full needs both. Label:** 🟡 `railway=*` + `waterway=*` co-located + vision. Restrict to rail+canal legacy cities. **Necessity:** Which infrastructure is the active severance at a neighborhood edge.

10. Land-use transition character — consensus 3/5 [recast directional → type labels]

Q: "This point sits on a mapped land-use boundary. What is the built-character change you actually see across it?" **Labels:** Residential → commercial/retail · Residential → industrial · Built-up → open/green · No visible change (boundary is administrative only). **Single-view but bad:** sat-only gives the boundary but residential vs light-industrial roofs look alike; sv-only reads character but doesn't know a boundary exists. **Full needs both. Label:** 🟡 edge between two `landuse` polygons + vision. **Needs polygon re-query.** ⚠️ correlation risk → sample administrative-boundary cases. **Necessity:** Zoning vs perceived boundary; district legibility.

TIER C — Useful, narrower or more clustered. Build if capacity allows.

11. Waterfront access type — conv 4/5 [recast → type labels]

Q: "A river/canal/coast is mapped near this point. What does its edge offer on the ground here?" **Labels:** Public promenade/path along the water · Road/bridge access to the water · Steps down to the water · No public access (walled/private/no route). **Label:** ● `waterway` / `coastline` + buffered `highway=footway/path` + vision. **Necessity:** Blue-space access, flood/heat-edge connectivity.

12. Nearest pedestrian crossing reachability — conv 2/5 [directional → text]

Q: "Crossing the road you're on, where is the nearest marked or signalized pedestrian crossing?" **Labels:** Just ahead along the road · Behind you along the road · Across to the left side · Across to the right side. **Label:** ● mostly Auto — `highway=crossing` nodes on the road + bearing → label. **Necessity:** Pedestrian safety / Vision Zero / accessibility for elderly, children, wheelchair users.

13. Transit access reachability — conv 3/5 [directional → text]

Q: "A transit stop/platform is mapped nearby. Where does a pedestrian go to reach it?" **Labels:** Ahead along this road · Back along this road · Onto the left cross-street · Onto the right cross-street. **Label:** ● `bus_stop` / `platform` projected to sidewalk + vision. **Necessity:** First/last-mile multimodal accessibility.

14. Cycleway continuity status — conv 1/5

Q: "The map tags this road as a cycleway. How does the physical cycle provision continue?" **Labels:** Continuous in both directions · Continues ahead only · Continues behind only · Discontinuous / signed-only (no physical lane). **Label:** ● `highway=cycleway` / `cycleway=*` + vision per frame. **Necessity:** Cycle-network continuity = top predictor of bike mode-share.

15. Arterial edge severance type — conv 2/5

Q: "This is a major arterial. How does its edge relate to the adjacent neighborhood?" **Labels:** Direct frontage/driveways onto the carriageway · Buffered by a parallel service road · Walled / sound-barrier severed · Open natural buffer. **Label:** ● `highway=primary/secondary` + parallel `service` / adjacency + vision. **Necessity:** Community severance, noise mitigation.

16. Mapped stairway approach — conv 3/5 [directional → text]

Q: "A mapped stairway is nearby. Where do you approach it from your road?" **Labels:** Straight ahead · Behind you · To the left · To the right. **Label:** ● Auto — `highway=steps` nearest + bearing (existence OSM-given; street confirms). **Necessity:** Accessibility (ramps vs stairs), emergency egress. Clustered to hilly cities.

17. Junction open-exit fraction — conv 2/5

Q: "The map shows several roads meeting here. How many are actually open to vehicles on the ground?" **Labels:** All of them · Most (at least half) · Few (less than half) · None / all blocked. **Label:** ● incident `highway` count + bearing-match + vision per exit. **Necessity:** "Paper street" / real vs mapped connectivity audit.

18. Pedestrian-crossing protection at a junction — conv 1/5

Q: "How does this junction's size relate to the pedestrian protection actually provided?" **Labels:** Compact and well-protected · Broad but with median refuges · Broad with no protection (stroad) · Physically severed (fenced, no at-grade crossing). **Label:** ● lane-count/area + vision. △ control footprint across classes. **Necessity:** Vision-Zero pedestrian safety.

19. Block cut-through presence — conv 3/5 [recast → type labels]

Q: "The map shows a dense building block at this point. What kind of pedestrian route through/around it does the street reveal?" **Labels:** An open passage cutting through the block · A gated/private passage (no public through) · Continuous building wall (no cut-through) · Only the perimeter sidewalk (no interior route). **Label:** 🟡 footpath attaching two roads, building-flanked, + vision. **Needs polygon re-query.** **Necessity:** Urban grain / permeability in dense cores.

20. Service/delivery access presence — conv 1/5 [recast → type labels]

Q: "What kind of vehicle access does the building/lot at this point have from the street?" **Labels:** Direct driveway/service entrance on this street · Access via a rear service lane · Parking-lot entrance · No vehicle access (pedestrian frontage only). **Label:** 🟡 `highway=service` into building/ `landuse=parking` polygon + vision. **Needs polygon re-query.** **Necessity:** Freight, curb, parking policy.

Recommended build order (for the meeting)

#1 (Forward link form) and #2 (Grade separation) first — top every model's ranking, most OSM-automatic (🟢, least vision), cleanest "single view is insufficient" argument, universal routing/severance need, and **no polygon re-query** (road graph over data you already fetch). Then **#3, #4, #5** for severance/connectivity breadth (also road-graph-only). Tier B/C add breadth but many need a one-time Overpass re-query (`landuse` / `leisure` /building polygons) + a vision-labelling pass.

Two rules for ALL 20 (non-negotiable)

1. **Balance answer labels ~25/25/25/25 WITHIN each city** — kills the satellite → city → modal-answer leak.
2. **Gate before scaling:** run `reasoning_distill/complementarity_gate.py` on a ~150-q pilot; require `full > each single by ≥8 pts` and `fusion-win ≥ 15%`. Single-view *may* be decent — it just must be clearly beaten. A task that fails the gate isn't fusion, however good the design argument sounds.

Honesty note for the team (the 🟡 tasks)

🟡 tasks use OSM to *select & filter* candidates but a VLM/human to assign the *eye-level answer label* (is it really stairs / a gate / an open edge?). Standard practice (UrBench, MMStar do this), but it means **labeling cost is real** and the benchmark is **curated, not fully auto-generated**. The most automatic (least vision) tasks — **#1, #2, #3, #12, #16** — are the safest "we can label at scale" bets.